## * structure

order

Different Data

flexable

* int
* float
* Double
* char
* string

Structures

* structures
* enum
* project

---

* Memory Managment

## * Structure

Different Data tyPe

struct

Char → 1 Byte

int → 4 Bytes

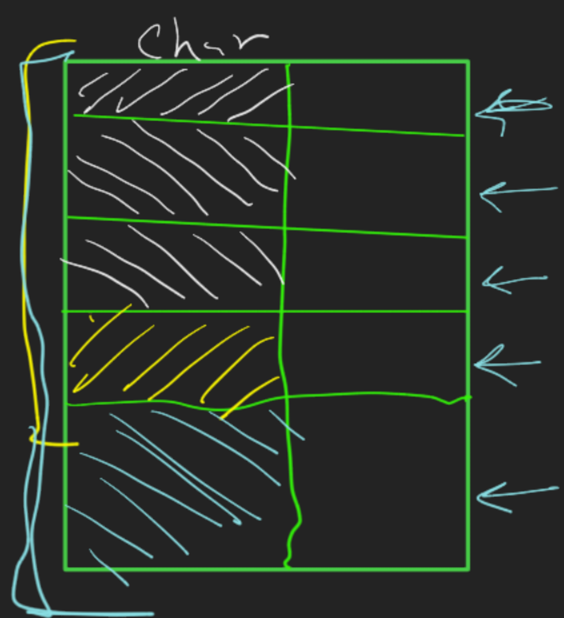
float → 4 Btye

Double → 8 Bytes

Array → Arr[5];

6

100




## Syntax

```
struct Namestruct {
    int MyAge
    char MyName[ ];
};
```

struct Namestruct S1;

# typedef

S1.MyAge = 25;

S1.MName[ ] = "Mahmoud Morsy"

→ var

13

12 1 1 \0

typedef

typedef int U16;

existed

Datatype

* structures pointers

```
typeDef struct {
    char brand[30];
    int year;
} Car;

int main() {

Car Car1 = {"Toyota", 2010};

Car *ptr = &Car1;
            └> Address

printf("year", ptr->year);
prinff("brand", ptr->brand);

}
```

* union

```
typedef union {
    int year;
    char bran[20];
}Car;
```

* struct

```
typedef struct {
    int year;
    char brand [20] Car
```

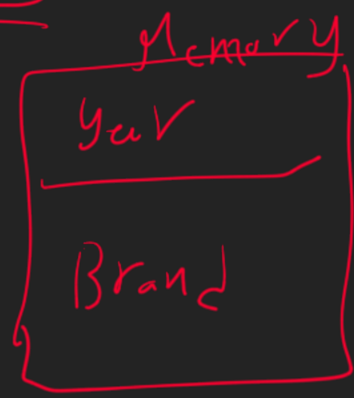
Memory

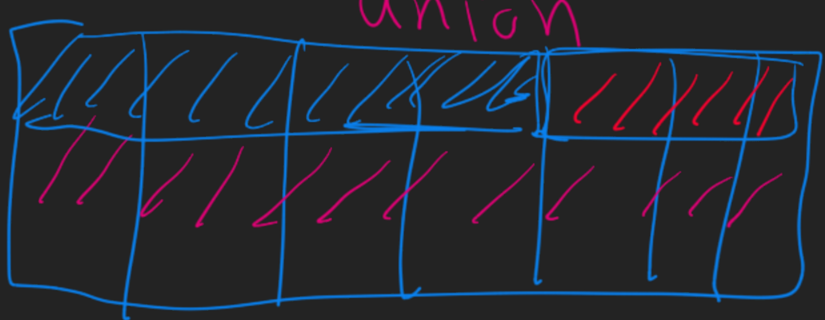
21 byte


20

16Byte

5 21 byte



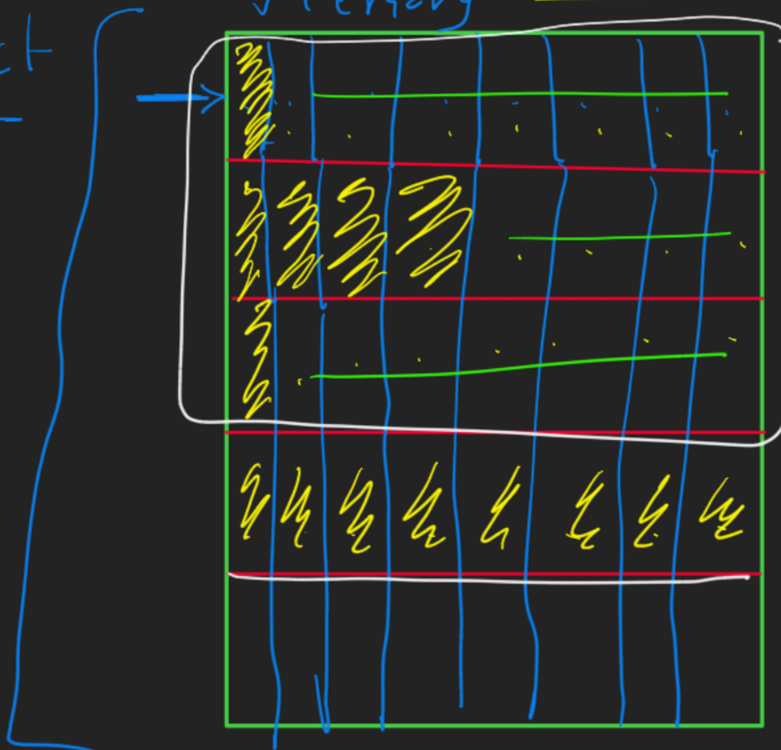

Largest Datatype

* padding

struct

Memory

فراغ



char → 1

int → 4

char → 1

double → 8

size of struct

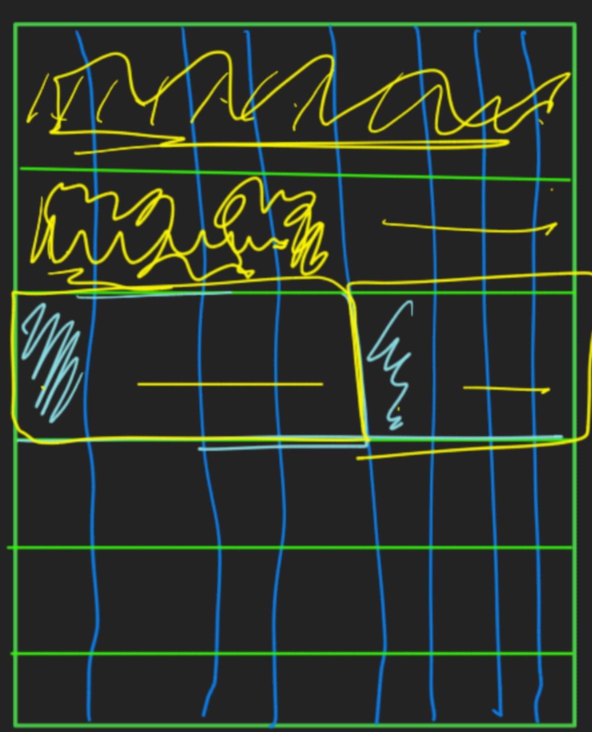
14 Byte

8 padding

double

int

char



char

14

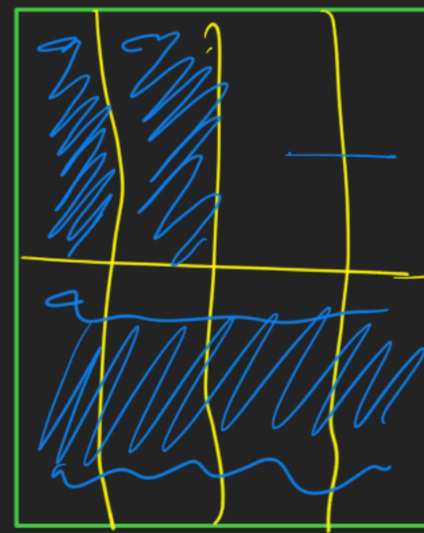
3 3 4

10

24

ad

ad

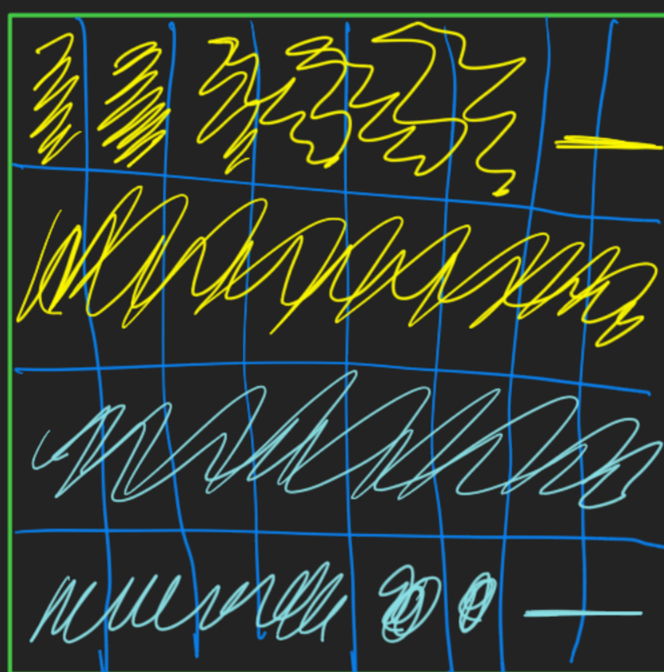


4

8

4

double



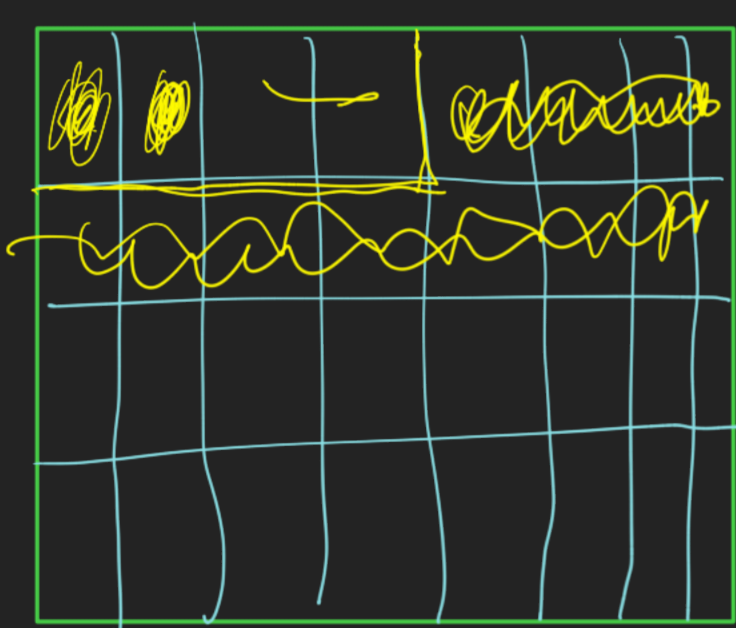
8

8

2

8

16



Datatypes

order

char
int
char
double

8

4